Testing Semantics for Quantum Distributed Systems

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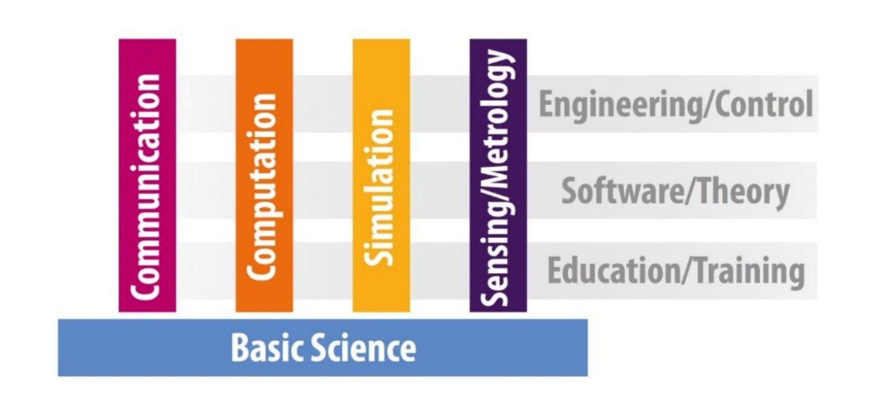
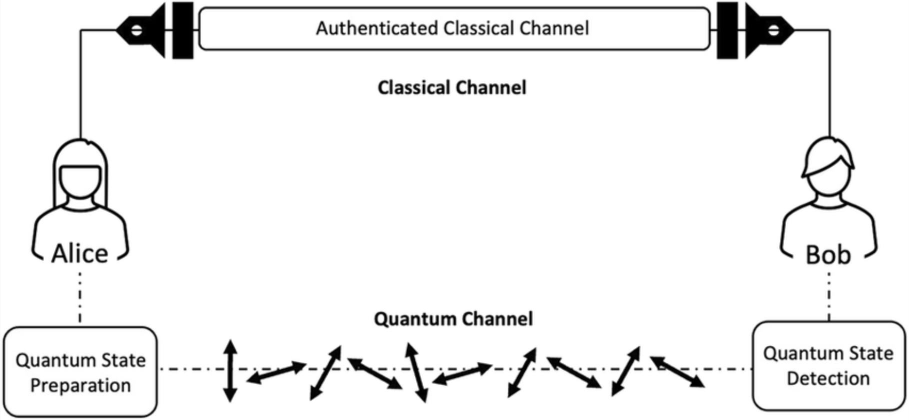
Quantum Technologies



Why Quantum Communication

- For Implementing **Quantum Algorithms**
  - speedup over classical counterparts
  - but computers with big registers are difficult
  - distributed computing with the quantum internet
- For **Quantum Protocols**
  - quantum key distribution; leader-election; superdense-coding
  - security guarantees
  - communication efficiency


Authenticated Classical Channel
Classical Channel
Alice
Bob
Quantum Channel
Quantum State Preparation
Quantum State Detection

Modeling and Verifying Quantum Distributed Systems

We need:

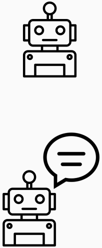
- Description language
- Semantic model
- Technique for checking correctness

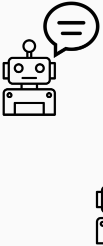
Process algebras have proven successful for modeling and verifying concurrent systems also with probabilities

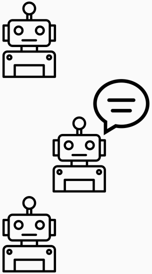
- We use them for modeling quantum concurrent systems
- We compare their behaviour using tests!

Recall — Testing Processes

Modelling and Comparing Concurrent Systems







Concurrent Processes + Communication Primitives + Nondeterministic Choices

Value Passing CCS

A language for concurrent, non-deterministic, communicating systems.

$$P ::= \mathbf{0} \mid \tau.P \mid c!e.P \mid c?x.P \mid P + P \mid P \setminus c \mid P \parallel P \mid \textbf{if } e \textbf{ then } P \textbf{ else } P$$

- $P + Q$ is the non-deterministic composition of P and Q
- $P \parallel Q$ is the parallel composition of P and Q
- $c?x.P$ receives a value on the channel c, $c!v.P$ sends the value v con channel c
- **if** v **then** P **else** Q behaves as P if $v = 0$, as Q if $v \neq 0$.

Operational Semantics

Labelled Transition system $\langle S, Act, \rightarrow \rangle$, with $\rightarrow \subseteq S \times Act \times S$

$$\frac{}{\tau.P \xrightarrow{\tau} P} \qquad \frac{e \Downarrow v}{c!e.P \xrightarrow{c!v} P} \qquad \frac{}{c?x.P \xrightarrow{c?v} P[v/x]} \qquad \frac{P \xrightarrow{\mu} P'}{P + Q \xrightarrow{\mu} P'} \qquad \frac{Q \xrightarrow{\mu} Q'}{P + Q \xrightarrow{\mu} Q'}$$

$$\frac{e \Downarrow 0}{\textbf{if } e \textbf{ then } P \textbf{ else } Q \xrightarrow{\mu} P} \qquad \frac{e \not\Downarrow 0}{\textbf{if } e \textbf{ then } P \textbf{ else } Q \xrightarrow{\mu} Q} \qquad \frac{P \xrightarrow{\mu} P' \quad \mu \neq c!v, c?v}{P \setminus c \xrightarrow{\mu} P' \setminus c}$$

$$\frac{P \xrightarrow{\mu} P'}{P \parallel Q \xrightarrow{\mu} P' \parallel Q} \qquad \frac{Q \xrightarrow{\mu} Q'}{P \parallel Q \xrightarrow{\mu} P \parallel Q'} \qquad \frac{P \xrightarrow{c?v} P' \quad Q \xrightarrow{c!v} Q'}{P \parallel Q \xrightarrow{\tau} P' \parallel Q'} \qquad \frac{P \xrightarrow{c!v} P' \quad Q \xrightarrow{c?v} Q'}{P \parallel Q \xrightarrow{\tau} P' \parallel Q'}$$

Example

$$P = \mathit{start}!0.\mathit{choice}?x.\textbf{if } x \textbf{ then } \mathit{out}!\mathit{coffee} \textbf{ else } \mathit{out}!\mathit{tea}$$

$$\begin{tikzcd}
 & P \arrow[d, "\mathit{start}!0"'] & \\
 & P' \arrow[dl, "\mathit{choice}?0"'] \arrow[dr, "\mathit{choice}?1"] & \\
P''[0/x] \arrow[d, "\mathit{out}!\mathit{coffee}"'] & & P''[1/x] \arrow[d, "\mathit{out}!\mathit{tea}"] \\
\mathbf{0} & & \mathbf{0}
\end{tikzcd}$$

Testing Equivalence in a Nutshell

How to verify that two processes are equivalent? With **Tests**.

Consider:

- The evolution of the processes under the same **test** T
- Tests are like processes with a distinct (successful) termination ω,

$$T ::= \omega \mid \mathbf{0} \mid \tau.T \mid c!v.T \mid c?x.T \mid T + T \mid \textbf{if } e \textbf{ then } T \textbf{ else } T$$

- Processes and tests evolve together $\langle P, T\rangle \xrightarrow{\tau} \langle P_1, T_1\rangle \xrightarrow{\tau} \dots$
- Two possible outcomes
  - $\dots \xrightarrow{\tau} \langle P_n, \omega\rangle$ — **The test is successful**
  - all other cases — **The test fails**

Test Semantics = Parallel Composition

$$\frac{P \xrightarrow{c!v} P'}{\langle\!\langle P, c?x.T \rangle\!\rangle \xrightarrow{\tau} \langle\!\langle P', T[{}^{v}/_{x}] \rangle\!\rangle} \qquad \frac{e \Downarrow v \quad P \xrightarrow{c?v} P'}{\langle\!\langle P, c!e.T \rangle\!\rangle \xrightarrow{\tau} \langle\!\langle P', T \rangle\!\rangle}$$

$$\frac{P \xrightarrow{\mu} P'}{\langle\!\langle P, T \rangle\!\rangle \xrightarrow{\mu} \langle\!\langle P', T \rangle\!\rangle} \qquad \frac{}{\langle\!\langle P, \tau.T \rangle\!\rangle \xrightarrow{\tau} \langle\!\langle P, T \rangle\!\rangle}$$

$$\frac{e \Downarrow 0}{\langle\!\langle P, \textbf{if } e \textbf{ then } T \textbf{ else } T' \rangle\!\rangle \xrightarrow{\tau} \langle\!\langle P, T \rangle\!\rangle} \qquad \frac{e \not\Downarrow 0}{\langle\!\langle P, \textbf{if } e \textbf{ then } T \textbf{ else } T' \rangle\!\rangle \xrightarrow{\tau} \langle\!\langle P, T' \rangle\!\rangle}$$

Testing Equivalence

Definition

A process P **may** pass the test T if $\langle P, T\rangle \xrightarrow{\tau}^{*} \langle P', \omega\rangle$ for some path.

A process P **must** pass the test T if $\langle P, T\rangle \xrightarrow{\tau}^{*} \langle P', \omega\rangle$ for all paths.

$$\begin{tikzcd} \langle a!0, a?x.\omega\rangle \arrow[d, "\tau"'] \\ \langle \mathbf{0}, \omega\rangle \end{tikzcd}$$

pass

$$\langle a!0, b?x.\omega\rangle$$

not pass

$$\begin{tikzcd} & \langle a!0 + a!1, a?x.\textbf{if } x \textbf{ then } \omega \textbf{ else } \mathbf{0}\rangle \arrow[dl, "\tau"'] \arrow[dr, "\tau"] & \\ \langle \mathbf{0}, \omega\rangle & & \langle \mathbf{0}, \mathbf{0}\rangle \end{tikzcd}$$

may pass, not must pass

Testing Equivalence

Definition
Two processes are **may-testing equivalent** if they may pass the same set of tests.

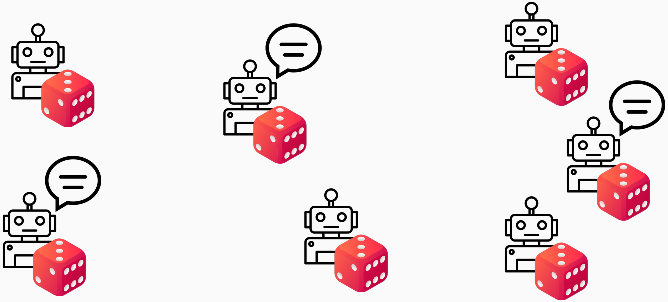
Two processes are **must-testing equivalent** if they must pass the same set of tests.

Two processes are **testing equivalent** if they are may and must testing equivalent.

$$
\begin{tikzcd}
& c!0 \parallel c!1 \arrow[dl, "c!0"'] \arrow[dr, "c!1"] & & & & c!0.c!1 + c!1.c!0 \arrow[dl, "c!0"'] \arrow[dr, "c!1"] & \\
\mathbf{0} \parallel c!1 \arrow[dr, "c!1"'] & & c!0 \parallel \mathbf{0} \arrow[dl, "c!0"] & \sim & c!1 \arrow[d, "c!1"'] & & c!0 \arrow[d, "c!0"] \\
& \mathbf{0} \parallel \mathbf{0} & & & \mathbf{0} & & \mathbf{0}
\end{tikzcd}
$$

Testing Probabilistic Processes

Modeling and Comparing Probabilistic Concurrent Systems



Concurrent Processes + Communication Primitives + Nondeterministic Choices + Random Sources

Probability Distributions

Finite **probability distributions** on X are functions from X to $[0,1]$

$$D(x) = \left\{ \Delta : X \to [0,1] \;\middle|\; \sum_{x \in X} \Delta(x) = 1,\ \lceil \Delta \rceil \text{ is finite} \right\}$$

where $\lceil \Delta \rceil = \{x \in X \mid \Delta(x) \neq 0\}$

Point distribution:

$$\overline{x}(y) = \begin{cases} 1 & \text{if } y = x \\ 0 & \text{otherwise} \end{cases} \qquad \ldots\text{we will often write } x \text{ for } \overline{x}$$

Convex combination:

$$(\Delta\ {}_p\!\oplus \Theta)(y) = p(\Delta(y)) + (1-p)(\Theta(y))$$

A Probabilistic Version of CCS

A language for concurrent, non-deterministic, and **probabilistic** communicating systems.

$$P ::= \mathbf{0} \mid \tau.P \mid c!v.P \mid c?x.P \mid P + P \mid P \setminus c \mid P \parallel P \mid \textbf{if } e \textbf{ then } P \textbf{ else } P \mid M_\Delta(x).P$$

- Δ is a probability distribution of natural numbers
- $M_\Delta(x)$ randomly selects an outcome from Δ and associates it to the variable x

Operational Semantics

Nondeterminist Probabilistic Labelled Transition system (NPLTS) $\langle S, Act, \rightarrow\rangle$, with $\rightarrow \subseteq S \times Act \times D(S)$

$$\frac{e \Downarrow v}{c!e.P \xrightarrow{c!v} \overline{P}} \qquad \frac{}{c?x.P \xrightarrow{c?v} \overline{P[v/x]}} \qquad \frac{P \xrightarrow{\mu} \Delta}{P + Q \xrightarrow{\mu} \Delta} \qquad \frac{Q \xrightarrow{\mu} \Theta}{P + Q \xrightarrow{\mu} \Theta}$$

$$\frac{}{\tau.P \xrightarrow{\tau} \overline{P}} \qquad \frac{e \Downarrow 0}{\textbf{if } e \textbf{ then } P \textbf{ else } Q \xrightarrow{\mu} \overline{P}} \qquad \frac{e \not\Downarrow 0}{\textbf{if } e \textbf{ then } P \textbf{ else } Q \xrightarrow{\mu} \overline{Q}}$$

Operational Semantics

Nondeterminist Probabilistic Labelled Transition system (NPLTS) $\langle S, Act, \rightarrow \rangle$, with $\rightarrow \subseteq S \times Act \times D(S)$

$$\frac{P \xrightarrow{\mu} \Delta \quad \mu \neq c!v, c?v}{P \setminus c \xrightarrow{\mu} \Delta \setminus c} \qquad \frac{P \xrightarrow{\mu} \Delta}{P \parallel Q \xrightarrow{\mu} \Delta \parallel Q} \qquad \frac{Q \xrightarrow{\mu} \Theta}{P \parallel Q \xrightarrow{\mu} P \parallel \Theta}$$

Where

$$(\Delta \setminus c)(P) = \begin{cases} \Delta(Q) & \text{if } P = Q \setminus c \\ 0 & \text{otherwise} \end{cases} \qquad (\Delta \parallel R)(P) = \begin{cases} \Delta(Q) & \text{if } P = Q \parallel R \\ 0 & \text{otherwise} \end{cases}$$

Operational Semantics

Nondeterminist Probabilistic Labelled Transition system (NPLTS) $\langle S, Act, \rightarrow \rangle$, with $\rightarrow \subseteq S \times Act \times D(S)$

$$\frac{P \xrightarrow{c?v} \overline{P} \quad Q \xrightarrow{c!v} \overline{Q}}{P \parallel Q \xrightarrow{\tau} \overline{P' \parallel Q'}} \qquad \frac{P \xrightarrow{c!v} \overline{P'} \quad Q \xrightarrow{c?v} \overline{Q'}}{P \parallel Q \xrightarrow{\tau} \overline{P' \parallel Q'}}$$

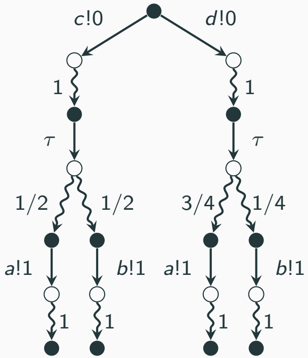
$$\frac{\Delta = \sum_{i \in I} p_i \cdot v_i}{M_\Delta(x).P \xrightarrow{\tau} \sum_{i \in I} p_i \cdot P[{}^{v_i}/_{x}]}$$

Example

$$c!0.M_{\mathsf{fair}}(x).\mathbf{if}\ x\ \mathbf{then}\ a!1\ \mathbf{else}\ b!1$$

$$+$$

$$d!0.M_{\mathsf{unfair}}(x).\mathbf{if}\ x\ \mathbf{then}\ a!1\ \mathbf{else}\ b!1$$



Testing

Distinguishing probabilistic processes

- As before:
  - the evolution of the process P in **parallel** with a **test** T
  - tests are like processes with a distinct (successful) termination ω
- This time: we must resolve **both non-determinism and probability!**

$$T ::= \mathbf{0} \mid \omega \mid \tau.T \mid c!v.T \mid c?x.T \mid T + T \mid \textbf{if } e \textbf{ then } T \textbf{ else } T \mid M_{\Delta}(x).P$$

Example

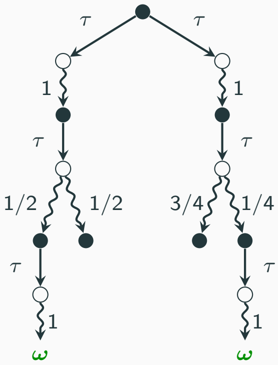
Process

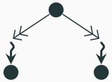
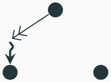
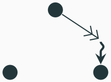
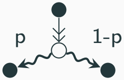
$$P \;:=\; c!0.M_{\mathsf{fair}}(x).\textbf{if } x \textbf{ then } a!1 \textbf{ else } b!1$$
$$+$$
$$d!0.M_{\mathsf{unfair}}(x).\textbf{if } x \textbf{ then } a!1 \textbf{ else } b!1$$

Test

$$T \;:=\; c?x.a?y.\boldsymbol{\omega}$$
$$+$$
$$d?x.b?y.\boldsymbol{\omega}$$

The evolution of $\langle P, T \rangle$

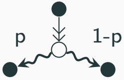


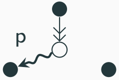
From  to  or  or 

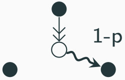
Definition (Resolution though randomized schedulers)
Given an NPTS $(S, \rightarrow)$, a *resolution* R is a PTS $(S, \rightarrow_R)$ such that for every $s \in S$

- if $s \rightarrow_R \Delta$ then there exists probabilities $\{p_i\}_{i \in I}$ and distributions $\{\Delta_i\}_{i \in I}$ such that $\sum_{i \in I} p_i = 1$, $\Delta = \sum_{i \in I} p_i \bullet \Delta_i$ and for each $i \in I$ there is a transition $s \rightarrow \Delta_i$ in the original NPTS.
- if $\nexists \Delta$ such that $s \rightarrow_R \Delta$, then $\nexists \Delta$ such that $s \rightarrow \Delta$ in the original NPTS.

Resolving Probability

From 

to 

or 

Definition (Computation)
Given P_0, T_0 and a resolution R, a computation of length n for $\langle P_0, T_0 \rangle$ is a sequence

$$c = \langle P_0, T_0 \rangle \xrightarrow{\tau}_R \langle P_1, T_1 \rangle, \cdots, \langle P_{n-1}, T_{n-1} \rangle \xrightarrow{\tau}_R \langle P_n, T_n \rangle$$

where, for $i = 1, \ldots, n$, $\langle P_i, T_i \rangle \in \lceil \Delta_i \rceil$ with Δ_i the unique distribution such that $\langle P_{i-1}, T_{i-1} \rangle \xrightarrow{\tau}_R \Delta_i$.

- We say that c is *maximal* if it is not a proper prefix of any other computation
- The *probability* of c is $prob(c) = \prod_{i=1}^{n} \Delta_i(P_i, T_i)$

Success Probability

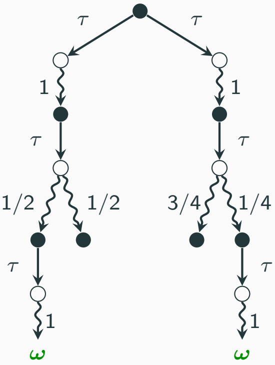
Definition (Success probability)
Given P, T and R,

$$sp_R(\langle\!|P, T|\!\rangle) = \sum_{c \in Succ_R(\langle\!|P,T|\!\rangle)} prob(c)$$

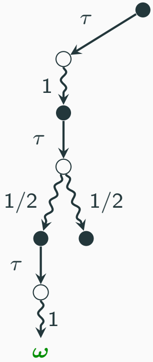
where $Succ_R(\langle\!|P, T|\!\rangle)$ is the set of maximal computations in R starting from $\langle\!|P, T|\!\rangle$ and containing a success state $\langle\!|P', \omega|\!\rangle$.

Example

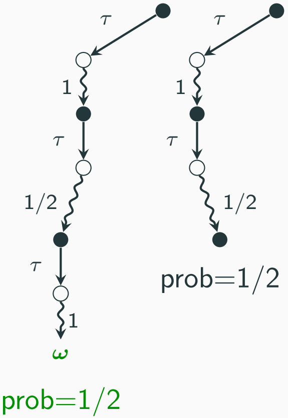
Evolution of $\langle P, T \rangle$



A resolution



Resolution's traces
(with probability)



Testing Equivalence

Definition (Testing Equivalence)
$P \sim_{\mathbb{T}} Q$, if for every test T,

- for each resolution R_1, there exists a resolution R_2 such that

$$sp_{R_1}(\langle\!| P, T |\!\rangle) = sp_{R_2}(\langle\!| Q, T |\!\rangle)$$

- for each resolution R_2, there exists a resolution R_1 such that

$$sp_{R_2}(\langle\!| Q, T |\!\rangle) = sp_{R_1}(\langle\!| P, T |\!\rangle)$$

Quantum Background

Hilbert Spaces

A (finite-dimensional) *Hilbert Space* $\mathcal{H}$ is a *complex vector space* with *inner product*.

- Column vector $|\psi\rangle$

$$|\psi\rangle = \begin{pmatrix} 0.6i \\ 0.2 + 0.2i \end{pmatrix}$$

- Conjugate transpose:

$$\langle\psi| = \begin{pmatrix} -0.6i & 0.2 - 0.2i \end{pmatrix}$$

- Dot product between $|\psi\rangle$ and $|\phi\rangle$:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad |\phi\rangle = \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \qquad \langle\psi|\phi\rangle = \begin{pmatrix} \overline{\alpha} & \overline{\beta} \end{pmatrix} \cdot \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$$

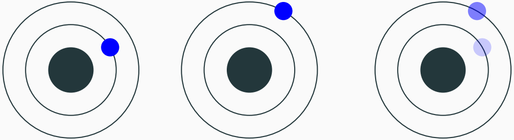
States of a Quantum System

A quantum state $|\phi\rangle$ is a unitary vector in a Hilbert space, i.e. $\langle\phi|\phi\rangle = 1$.

For **bits**: two *classical states* 0 and 1

A **qubit** may be in a *superposition* of the two

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \qquad \text{with amplitudes } \alpha, \beta \in \mathbb{C} \text{ such that } |\alpha|^2 + |\beta|^2 = 1$$



$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \tfrac{1}{2}\,|0\rangle + \tfrac{\sqrt{3}}{2}\,|1\rangle$$

Basis

Computational basis: $B_{01} = \{|0\rangle, |1\rangle\}$.

Hadamard basis: $B_{\pm} = \{|+\rangle, |-\rangle\}$ with

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Imaginary basis: $B_{\pm i} = \{|i\rangle, |-i\rangle\}$ with

$$|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) \qquad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$$

Programmers use Unitary Transformations to Change the Qubits State

A Couple of Examples

Quantum version of bit flip

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$X|0\rangle = |1\rangle$$

$$X|1\rangle = |0\rangle$$

Basis mapping $B_{01} \Longleftrightarrow B_{\pm}$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = |+\rangle$$

$$H|1\rangle = |-\rangle$$

$$H|+\rangle = |0\rangle$$

$$H|-\rangle = |1\rangle$$

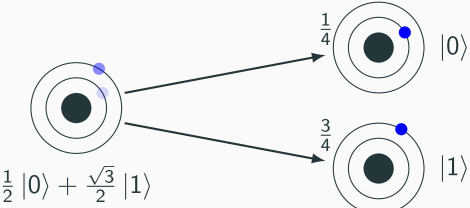
Projective Measurements

Measurement on a given basis: $M_{01}, M_{\pm}$ with a probabilistic result composed by

- the **classical outcome**
- how the **state decays**

For M_{01}, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

- 0 and $|0\rangle$ with probability $|\alpha|^2$
- 1 and $|1\rangle$ with probability $|\beta|^2$



Qubits cannot be observed without affecting their state!

Projective Measurements for our Bases

Our bases are all incompatible!: $M_{01}, M_{\pm}, M_{\pm i}$

- measuring $|0\rangle$ in $\{|0\rangle, |1\rangle\}$ always gives $|0\rangle$
- measuring $|0\rangle$ in $\{|+\rangle, |-\rangle\}$ always gives $|+\rangle \;{}_{1/2}\!\oplus |-\rangle$
- measuring $|+\rangle$ in $\{|+\rangle, |-\rangle\}$ always gives $|+\rangle$
- measuring $|+\rangle$ in $\{|0\rangle, |1\rangle\}$ always gives $|0\rangle \;{}_{1/2}\!\oplus |1\rangle$

A Remark on Measurement

Assume $|\psi\rangle$ is one of $|0\rangle, |1\rangle, |+\rangle, |-\rangle$.

How can you know which one?

- You can try with M_{01}, but maybe $|\psi\rangle$ is in $\{|+\rangle, |-\rangle\}$
- You can try with $M_{\pm}$, but maybe $|\psi\rangle$ is in $\{|0\rangle, |1\rangle\}$

In both cases you may get useless information from the measurement and **destroy** the original state of the qubit

Measurement cannot discriminate with arbitrary precision!

Composite Quantum Systems

States and transformations composed through *tensor product*, or *kronecker product*.

$$A \otimes B = \begin{pmatrix} a_{11}B & \dots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \dots & a_{nn}B \end{pmatrix}$$

$$|0\rangle \otimes |0\rangle = |00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$|+\rangle \otimes |0\rangle = |+0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

A transformation X applied on just the first qubit is $X \otimes I =$

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

No-Cloning Theorem

Theorem. There is no unitary transformation U and state $|\psi\rangle$ such that for every $|\phi\rangle$

$$U(|\phi\rangle \otimes |\psi\rangle) = |\phi\rangle \otimes |\phi\rangle$$

No broadcasting!

Entanglement

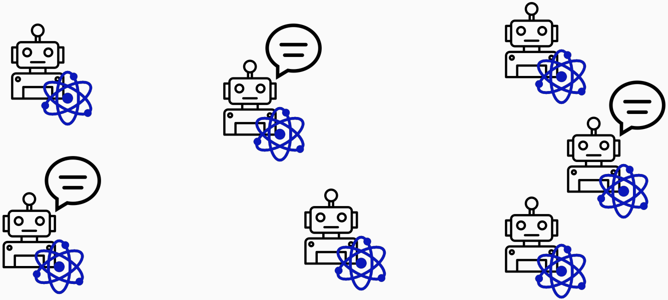
A state that cannot be the product of two smaller states

Definition. $|\psi\rangle$ is entangled iff $\forall\, |\phi_1\rangle, |\phi_2\rangle \,.\, |\psi\rangle \neq |\phi_1\rangle \otimes |\phi_2\rangle$

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \implies \begin{array}{l} M_{01}(|\Phi^+\rangle) = |00\rangle \;{}_{1/2}{\oplus}\; |11\rangle \\ M_{\pm}(|\Phi^+\rangle) = |{+}{+}\rangle \;{}_{1/2}{\oplus}\; |{-}{-}\rangle \end{array}$$

A Quantum Process Algebra

Modelling and Comparing Quantum Concurrent Systems



Concurrent Processes + Communication Primitives + Nondeterministic Choices + Quantum Capable

lqCCS

$$P ::= \mathbf{0}_{\tilde{q}} \mid \tau.P \mid c!e.P \mid c?x.P \mid P + P \mid P \setminus c \mid P \parallel P \mid \textbf{if } e \textbf{ then } P \textbf{ else } P$$
$$\mid U(\tilde{e}).P \mid M_B(\tilde{e} \triangleright x).P$$

where U is a unitary transformation, M_B is a measurement on the basis B, and e is an expression over numbers or qubit names $q_1, q_2, \ldots$.

The semantics of $\langle\!|\, |\psi\rangle, P \,|\!\rangle \in \mathit{Conf}$ is a **NPLTS**

Nondeterministic **P**robabilistic **L**abelled **T**ransition **S**ystem

$$\langle \mathit{Conf}, \mathit{Act}, \rightarrow \subseteq \mathit{Conf} \times \mathit{Act} \times \mathcal{D}(\mathit{Conf}) \rangle$$

Problem with Cloning Qubits

$$
\begin{array}{c}
\langle |0\rangle\,, c!q_1.d!q_1 \parallel c?x.P \parallel d?x.Q \rangle \\
\tau \big\downarrow \\
\circ \\
\rightsquigarrow \\
\langle |0\rangle\,, d!q_1 \parallel P[{}^{q_1}\!/_{x}] \parallel d?x.Q \rangle \\
\tau \big\downarrow \\
\circ \\
\rightsquigarrow \\
\langle |0\rangle\,, \mathbf{0} \parallel P[{}^{q_1}\!/_{x}] \parallel Q[{}^{q_1}\!/_{x}] \rangle
\end{array}
$$

This process is not physically implementable

- Sends q_1 along both c and d
- Requires copying the qubit state
- Contradicts the no-cloning theorem

Linear Type System for Qubits Names

Linearity: Single ownership of qubits (implies *no-cloning theorem*)

- No qubit is shared by more processes in parallel
- At the end, each qubit is either sent with $c!q$ or discarded with $\mathbf{0}_q$
- Operations are performed only on owned qubits

Enforced via a **linear type system**

$$\Sigma \vdash P \qquad \text{with } \Sigma \text{ a set of qubit names}$$

Means that P is correct and uses all (and only) the qubits in Σ

Linear Type System for Qubits Names

$$\frac{\tilde{e} \in \tilde{\Sigma}}{\Sigma \vdash \mathbf{0}_{\tilde{e}}} \qquad \frac{\Sigma \vdash P}{\Sigma \vdash \tau.P} \qquad \frac{U : \mathrm{Op}(n) \quad |E| = n \quad \tilde{e} \in \tilde{E} \quad E \subseteq \Sigma \quad \Sigma \vdash P}{\Sigma \vdash U(\tilde{e}).P} \qquad \frac{\Sigma \vdash P}{\Sigma \vdash P \setminus c}$$

$$\frac{M_B : \mathrm{Meas}(n) \quad |E| = n \quad \tilde{e} \in \tilde{E} \quad E \subseteq \Sigma \quad x : \mathbb{N} \quad \Sigma \vdash P}{\Sigma \vdash M_B(\tilde{e} \triangleright x).P}$$

$$\frac{c : \widehat{\mathbb{N}} \quad x : \mathbb{N} \quad \Sigma \vdash P}{\Sigma \vdash c?x.P} \qquad \frac{c : \widehat{\mathcal{Q}} \quad x : \mathcal{Q} \quad \Sigma \cup \{x\} \vdash P}{\Sigma \vdash c?x.P}$$

$$\frac{c : \widehat{\mathbb{N}} \quad e : \mathbb{N} \quad \Sigma \vdash P}{\Sigma \vdash c!e.P} \qquad \frac{c : \widehat{\mathcal{Q}} \quad e \in \Sigma \quad \Sigma \setminus \{e\} \vdash P}{\Sigma \vdash c!e.P}$$

$$\frac{e : \mathbb{B} \quad \Sigma \vdash P \quad \Sigma \vdash Q}{\Sigma \vdash \textbf{if } e \textbf{ then } P \textbf{ else } Q} \quad \frac{\Sigma \vdash P \quad \Sigma \vdash Q}{\Sigma \vdash P + Q} \quad \frac{\Sigma_1 \cap \Sigma_2 = \emptyset \quad \Sigma_1 \vdash P \quad \Sigma_2 \vdash Q}{\Sigma_1 \cup \Sigma_2 \vdash P \parallel Q}$$

Linear Type System for Qubits Names

Examples:

$$\{q_1\} \vdash c?x.H(x).X(q_1).(d!q_1 \parallel e!x)$$

$$\nvdash c?x.H(x).(d!x \parallel H(x).c!x)$$

$$\{q_1, q_2\} \vdash M_B(q_1 \triangleright x).\textbf{if } x \textbf{ then } c!q_1.\mathbf{0}_{q_2} \textbf{ else } c!q_2.\mathbf{0}_{q_1}$$

- **Theorem:** typing contexts are unique
- Hereafter, we work only with well-typed processes
- We write Σ_P for the only Σ such that $\Sigma \vdash P$
- We also write $\Sigma_{|\psi\rangle}$ for the qubits appearing in the state $|\psi\rangle$
- A configuration $\langle\!\langle |\psi\rangle, P \rangle\!\rangle \in \mathit{Conf}$ is well typed if $\Sigma_P \subseteq \Sigma_{|\psi\rangle}$
- Hereafter, we work only with well-typed configurations

Operational Semantics

The classical fragment... is quite standard

$$\frac{e \Downarrow 0}{\langle\!| \lvert\psi\rangle, \textbf{if } e \textbf{ then } P \textbf{ else } Q |\!\rangle \xrightarrow{\tau} \overline{\langle\!| \lvert\psi\rangle, P |\!\rangle}} \qquad \frac{e \not\Downarrow 0}{\langle\!| \lvert\psi\rangle, \textbf{if } e \textbf{ then } P \textbf{ else } Q |\!\rangle \xrightarrow{\tau} \overline{\langle\!| \lvert\psi\rangle, Q |\!\rangle}}$$

$$\frac{e \Downarrow v}{\langle\!| \lvert\psi\rangle, c!e.P |\!\rangle \xrightarrow{c!v} \overline{\langle\!| \lvert\psi\rangle, P |\!\rangle}} \qquad \frac{v \in \Sigma_{\lvert\psi\rangle} \setminus \Sigma_P}{\langle\!| \lvert\psi\rangle, c?x.P |\!\rangle \xrightarrow{c?v} \overline{\langle\!| \lvert\psi\rangle, P[v/x] |\!\rangle}}$$

$$\frac{\langle\!| \lvert\psi\rangle, P |\!\rangle \xrightarrow{\mu} \Delta}{\langle\!| \lvert\psi\rangle, P + Q |\!\rangle \xrightarrow{\mu} \Delta} \quad \frac{\langle\!| \lvert\psi\rangle, Q |\!\rangle \xrightarrow{\mu} \Theta}{\langle\!| \lvert\psi\rangle, P + Q |\!\rangle \xrightarrow{\mu} \Theta} \qquad \frac{}{\langle\!| \lvert\psi\rangle, \tau.P |\!\rangle \xrightarrow{\tau} \overline{\langle\!| \lvert\psi\rangle, P |\!\rangle}}$$

Operational Semantics

The classical fragment... is quite standard

$$\frac{\langle\!|\, |\psi\rangle\,, P \,|\!\rangle \xrightarrow{\mu} \Delta \quad \mu \neq c!v, c?v}{\langle\!|\, |\psi\rangle\,, P \setminus c \,|\!\rangle \xrightarrow{\mu} \Delta \setminus c}$$

$$\frac{\langle\!|\, |\psi\rangle\,, P \,|\!\rangle \xrightarrow{\mu} \Delta \quad \mu = c?v \Rightarrow v \notin \Sigma_Q}{\langle\!|\, |\psi\rangle\,, P \parallel Q \,|\!\rangle \xrightarrow{\mu} \Delta \parallel Q}$$

$$\frac{\langle\!|\, |\psi\rangle, P \,|\!\rangle \xrightarrow{c!v} \overline{\langle\!|\, |\psi\rangle, P' \,|\!\rangle} \quad \langle\!|\, |\psi\rangle, Q \,|\!\rangle \xrightarrow{c?v} \overline{\langle\!|\, |\psi\rangle, Q' \,|\!\rangle}}{\langle\!|\, |\psi\rangle, P \parallel Q \,|\!\rangle \xrightarrow{\tau} \overline{\langle\!|\, |\psi\rangle, P' \parallel Q' \,|\!\rangle}}$$

$$\frac{\langle\!|\, |\psi\rangle\,, Q \,|\!\rangle \xrightarrow{\mu} \Theta \quad \mu = c?v \Rightarrow v \notin \Sigma_P}{\langle\!|\, |\psi\rangle\,, P \parallel Q \,|\!\rangle \xrightarrow{\mu} P \parallel \Theta}$$

$$\frac{\langle\!|\, |\psi\rangle, P \,|\!\rangle \xrightarrow{c?v} \overline{\langle\!|\, |\psi\rangle, P' \,|\!\rangle} \quad \langle\!|\, |\psi\rangle, Q \,|\!\rangle \xrightarrow{c!v} \overline{\langle\!|\, |\psi\rangle, Q' \,|\!\rangle}}{\langle\!|\, |\psi\rangle, P \parallel Q \,|\!\rangle \xrightarrow{\tau} \overline{\langle\!|\, |\psi\rangle, P' \parallel Q' \,|\!\rangle}}$$

Operational Semantics

We add specific rules for quantum operations:

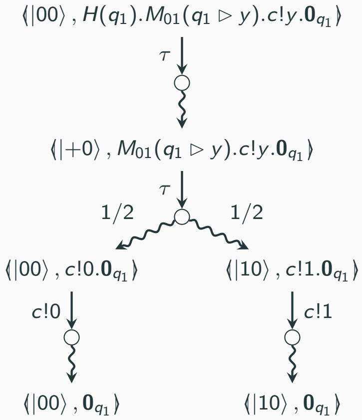
$$\frac{}{\langle\!\langle |\psi\rangle, U(\tilde{q}).P \rangle\!\rangle \xrightarrow{\tau} \overline{\langle\!\langle U^{\tilde{q}}|\psi\rangle, P \rangle\!\rangle}}$$

$$\frac{}{\langle\!\langle |\psi\rangle, M_B(\tilde{q} \triangleright x).P \rangle\!\rangle \xrightarrow{\tau} \sum_{m=0}^{|B|-1} \langle\psi|M_m^{\tilde{q}}|\psi\rangle \overline{\left\langle\!\!\left\langle \frac{M_m^{\tilde{q}}|\psi\rangle}{\langle\psi|M_m^{\tilde{q}}|\psi\rangle}, P[{}^{m}/_{x}] \right\rangle\!\!\right\rangle}}$$

where $L^{\tilde{q}}$ is a tensor composition of

- L on the qubits of $\tilde{q}$, and
- the identity on all the other qubits of $\Sigma_{|\psi\rangle}$

For Example


$\langle |00\rangle , H(q_1).M_{01}(q_1 \triangleright y).c!y.\mathbf{0}_{q_1} \rangle$
τ
$\langle |{+}0\rangle , M_{01}(q_1 \triangleright y).c!y.\mathbf{0}_{q_1} \rangle$
τ
$1/2$
$1/2$
$\langle |00\rangle , c!0.\mathbf{0}_{q_1} \rangle$
$\langle |10\rangle , c!1.\mathbf{0}_{q_1} \rangle$
$c!0$
$c!1$
$\langle |00\rangle , \mathbf{0}_{q_1} \rangle$
$\langle |10\rangle , \mathbf{0}_{q_1} \rangle$

Tests

Tests

Tests $\mathbb{T}_G$ are defined as

$$T \coloneqq \boldsymbol{\omega} \mid \mathbf{0} \mid \textbf{if } e \textbf{ then } T \textbf{ else } T \mid c?x.T \mid c!e.T \mid T + T \mid U(\tilde{e}).T \mid M(\tilde{e} \triangleright x).T$$

The semantics of a lqCCS extended configuration $\langle\!\langle |\psi\rangle, P, T \rangle\!\rangle \in TConf$ is a

Non-deterministic **P**robabilistic **T**ransition **S**ystem (**NPTS**)

$$\mathcal{T} = (TConf, \rightarrow \subseteq TConf \times \mathcal{D}(TConf))$$

Testing Equivalence

Definition (Testing Equivalence)
$\langle\!\langle |\psi\rangle, P \rangle\!\rangle \sim_{\mathbb{T}} \langle\!\langle |\phi\rangle, Q \rangle\!\rangle$, if for every test $T \in \mathbb{T}$,

- for each resolution R_1, there exists a resolution R_2 such that

$$sp_{R_1}(\langle\!\langle |\psi\rangle, P, T \rangle\!\rangle) = sp_{R_2}(\langle\!\langle |\phi\rangle, Q, T \rangle\!\rangle)$$

- for each resolution R_2, there exists a resolution R_1 such that

$$sp_{R_2}(\langle\!\langle |\phi\rangle, Q, T \rangle\!\rangle) = sp_{R_1}(\langle\!\langle |\psi\rangle, P, T \rangle\!\rangle)$$

Example: Quantum Lottery

State: $|0\rangle$

Process: $\mathbf{QL} = \mathit{Pre} \parallel \mathit{Ann}$

$$\mathbf{Pre} = (X(q).c!q.\mathbf{0}) + (H(q).c!q.\mathbf{0})$$

$$\mathbf{Ann} = c?x.M_{01}(x \triangleright y).\mathbf{if}\ y\ \mathbf{then}\ a!1.\mathbf{0}_x\ \mathbf{else}\ b!1.\mathbf{0}_x$$

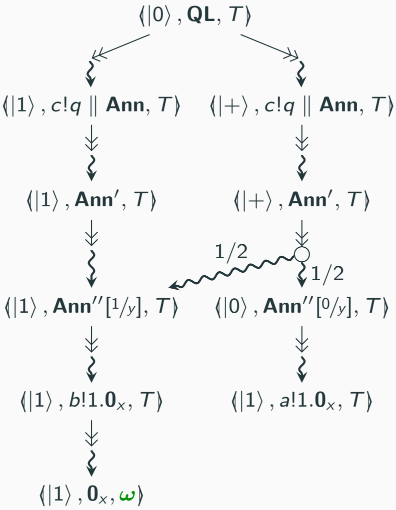
Test: $T = b?x.\boldsymbol{\omega}$

- **Pre** prepares a qubit used as a source of randomness
- **Ann** receives and measure it, and announces the winner, Alice $a!1$ or Bob $b!1$
- The test is successful if Bob wins the lottery

Example: Quantum Lottery Semantics (NPTS) — and a Resolution (PTS)

$$
\begin{tikzcd}[column sep=small]
 & \langle \lvert 0\rangle, \mathbf{QL}, T \rangle \arrow[dl, two heads] \arrow[dr, two heads] & \\
{} \arrow[d, rightsquigarrow] & & {} \arrow[d, rightsquigarrow] \\
\langle \lvert 1\rangle, c!q \parallel \mathbf{Ann}, T \rangle \arrow[d, two heads] & & \langle \lvert +\rangle, c!q \parallel \mathbf{Ann}, T \rangle \arrow[d, two heads] \\
{} \arrow[d, rightsquigarrow] & & {} \arrow[d, rightsquigarrow] \\
\langle \lvert 1\rangle, \mathbf{Ann}', T \rangle \arrow[d, two heads] & & \langle \lvert +\rangle, \mathbf{Ann}', T \rangle \arrow[d, two heads] \\
{} \arrow[d, rightsquigarrow] & & \circ \arrow[dll, rightsquigarrow, "1/2"'] \arrow[d, rightsquigarrow, "1/2"] \\
\langle \lvert 1\rangle, \mathbf{Ann}''[1/y], T \rangle \arrow[d, two heads] & & \langle \lvert 0\rangle, \mathbf{Ann}''[0/y], T \rangle \arrow[d, two heads] \\
{} \arrow[d, rightsquigarrow] & & {} \arrow[d, rightsquigarrow] \\
\langle \lvert 1\rangle, b!1.\mathbf{0}_x, T \rangle \arrow[d, two heads] & & \langle \lvert 1\rangle, a!1.\mathbf{0}_x, T \rangle \\
{} \arrow[d, rightsquigarrow] & & \\
\langle \lvert 1\rangle, \mathbf{0}_x, \boldsymbol{\omega} \rangle & &
\end{tikzcd}
$$

Example: Quantum Lottery Semantics (NPTS) — and a Resolution (PTS)



Ann

$c?x.M_{01}(x \triangleright y).\mathbf{if}\ y\ \mathbf{then}\ a!1.\mathbf{0}_x\ \mathbf{else}\ b!1.\mathbf{0}_x$

Ann'

$M_{01}(x \triangleright y).\mathbf{if}\ y\ \mathbf{then}\ a!1.\mathbf{0}_x\ \mathbf{else}\ b!1.\mathbf{0}_x$

Ann"

$\mathbf{if}\ y\ \mathbf{then}\ a!1.\mathbf{0}_x\ \mathbf{else}\ b!1.\mathbf{0}_x$

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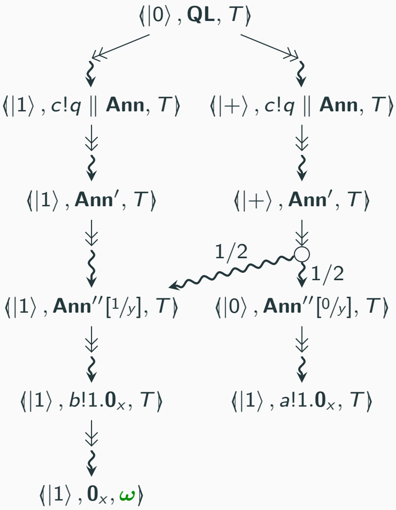

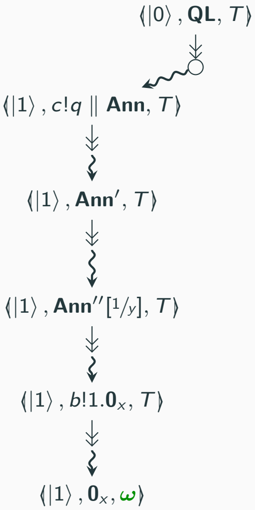

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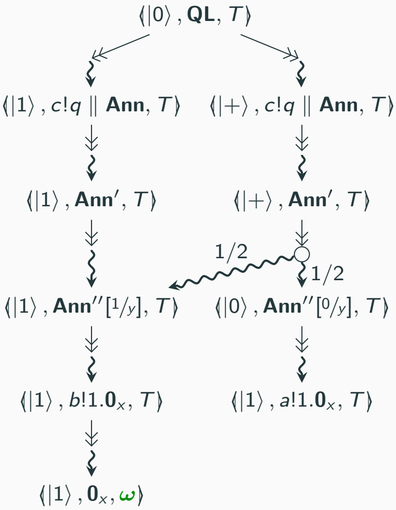
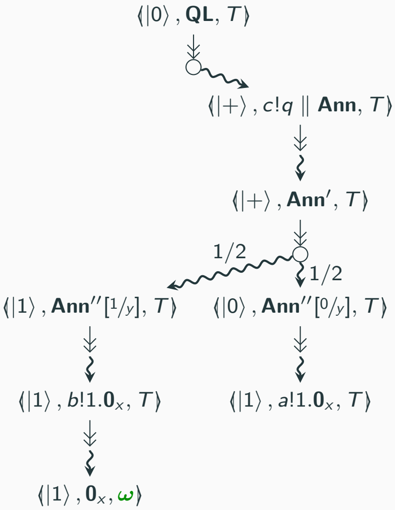
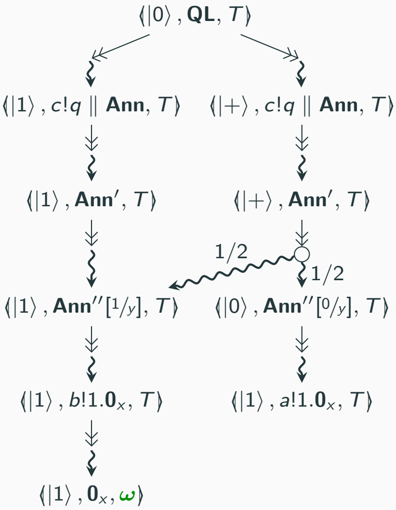
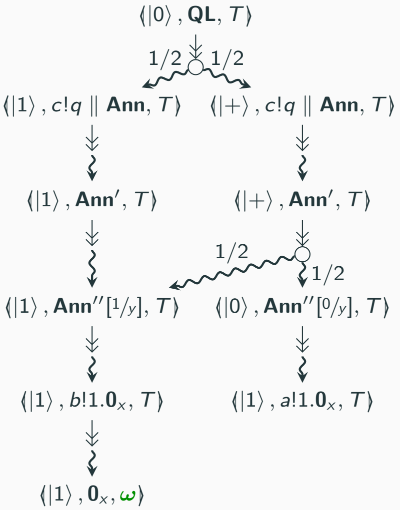




Example: Quantum Lottery Semantics (NPTS) — and a Resolution (PTS)

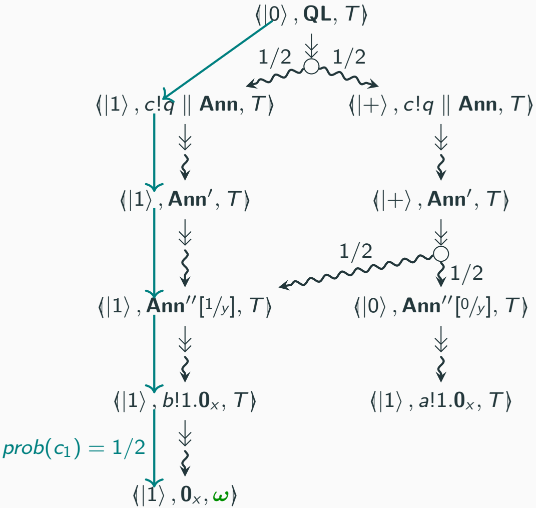


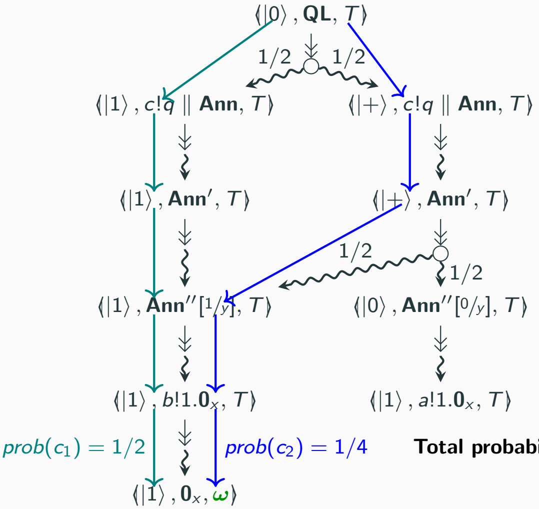


Example: Quantum Lottery Resolution (PTS) and its Computations

$$
\begin{tikzcd}[column sep=small]
 & \langle |0\rangle, \mathbf{QL}, T \rangle \arrow[dl, squiggly, "\twoheaddownarrow\ 1/2"'] \arrow[dr, squiggly, "1/2"] & \\
\langle |1\rangle, c!q \parallel \mathbf{Ann}, T \rangle \arrow[d, squiggly, "\twoheaddownarrow"'] & & \langle |+\rangle, c!q \parallel \mathbf{Ann}, T \rangle \arrow[d, squiggly, "\twoheaddownarrow"'] \\
\langle |1\rangle, \mathbf{Ann}', T \rangle \arrow[d, squiggly, "\twoheaddownarrow"'] & & \langle |+\rangle, \mathbf{Ann}', T \rangle \arrow[dll, squiggly, "\twoheaddownarrow\ 1/2"'] \arrow[d, squiggly, "1/2"] \\
\langle |1\rangle, \mathbf{Ann}''[{}^{1}/_{y}], T \rangle \arrow[d, squiggly, "\twoheaddownarrow"'] & & \langle |0\rangle, \mathbf{Ann}''[{}^{0}/_{y}], T \rangle \arrow[d, squiggly, "\twoheaddownarrow"'] \\
\langle |1\rangle, b!1.\mathbf{0}_x, T \rangle \arrow[d, squiggly, "\twoheaddownarrow"'] & & \langle |1\rangle, a!1.\mathbf{0}_x, T \rangle \\
\langle |1\rangle, \mathbf{0}_x, \boldsymbol{\omega} \rangle & &
\end{tikzcd}
$$

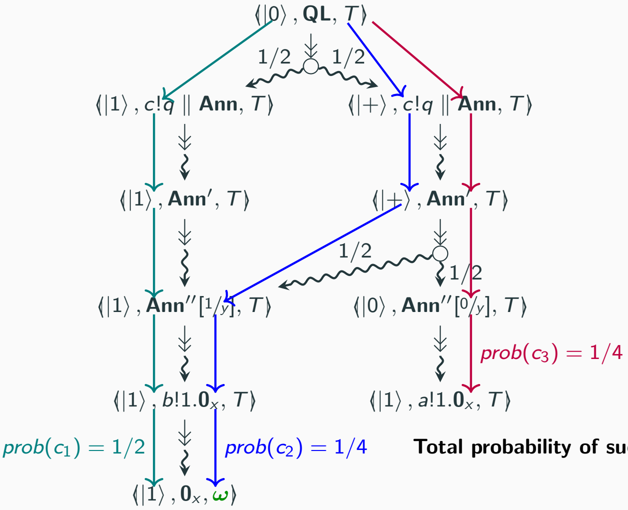
Example: Quantum Lottery Resolution (PTS) and its Computations





Total probability of success $3/4$

Example: Quantum Lottery Resolution (PTS) and its Computations



Total probability of success 3/4

The Problem of Non-deterministic Testing

A Missing Expected Equivalence

Take a pair of non-biased random qubit sources

- the first sends $|0\rangle$ or $|1\rangle$ (both with probability $1/2$)
- the second sends $|+\rangle$ or $|-\rangle$ (both with probability $1/2$)

Quantum theory prescribes that they cannot be distinguished by any observer, as the received qubits behave the same...

But they are distinguished by a non-deterministic test!

Formalizing the Counterexample

The two qubit sources in lqCCS

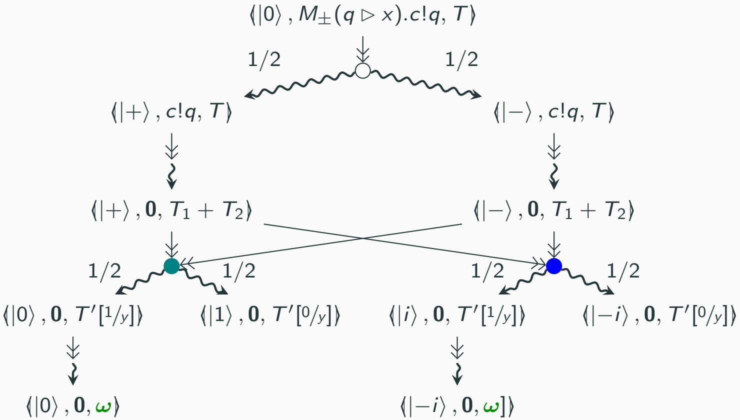
$$C_{01} = \langle\!\langle |+\rangle , M_{01}(q \triangleright x).c!q \rangle\!\rangle \quad \text{and} \quad C_{\pm} = \langle\!\langle |0\rangle , M_{\pm}(q \triangleright x).c!q \rangle\!\rangle$$

The distinguishing test $T = c?x.(T_1 + T_2)$ with

$$\begin{aligned} T_1 &= M_{01}(x \triangleright y).\mathbf{if}\ y\ \mathbf{then}\ \omega\ \mathbf{else}\ \mathbf{0}, \text{ and} \\ T_2 &= M_{\pm i}(x \triangleright y).\mathbf{if}\ y\ \mathbf{then}\ \omega\ \mathbf{else}\ \mathbf{0}. \end{aligned}$$

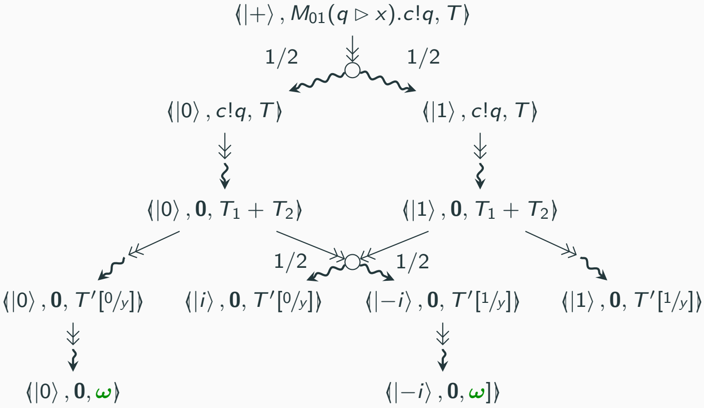
where $M_{\pm i}$ stands for the measurement $\{|i\rangle , |-i\rangle\}$.

Testing the Source of $|+\rangle$ and $|-\rangle$



Choosing teal or blue is the same: for any resolution, success probability is $1/2$

Testing the Source of $|0\rangle$ and $|1\rangle$



For this resolution, the probability of success is 3/4!

Testing the Source of $|0\rangle$ and $|1\rangle$

$$\langle\!\langle |+\rangle, M_{01}(q \triangleright x).c!q, T \rangle\!\rangle$$

$$\Downarrow \quad \tfrac{1}{2} \;\; \tfrac{1}{2}$$

$$\langle\!\langle |0\rangle, c!q, T \rangle\!\rangle \qquad \langle\!\langle |1\rangle, c!q, T \rangle\!\rangle$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$\langle\!\langle |0\rangle, \mathbf{0}, T_1 + T_2 \rangle\!\rangle \qquad \langle\!\langle |1\rangle, \mathbf{0}, T_1 + T_2 \rangle\!\rangle$$

$$\tfrac{1}{2} \;\; \tfrac{1}{2}$$

$$\langle\!\langle |0\rangle, \mathbf{0}, T'[{}^0/_y] \rangle\!\rangle \qquad \langle\!\langle |i\rangle, \mathbf{0}, T'[{}^0/_y] \rangle\!\rangle \qquad \langle\!\langle |-i\rangle, \mathbf{0}, T'[{}^1/_y] \rangle\!\rangle \qquad \langle\!\langle |1\rangle, \mathbf{0}, T'[{}^1/_y] \rangle\!\rangle$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$\langle\!\langle |0\rangle, \mathbf{0}, \boldsymbol{\omega} \rangle\!\rangle \qquad \langle\!\langle |-i\rangle, \mathbf{0}, \boldsymbol{\omega}] \rangle\!\rangle$$

For this resolution, the probability of success is $3/4$!

Why?

$$C_{01} \not\sim_{\mathbb{T}_G} C_{\pm}$$

- The reason of higher success for C_{01} is that we have chosen the measurement based on the quantum state of the received qubit
- But how do we know the state of the received qubit?
- Usually through a measurement... but we did not measure the qubit
- Being capable of inspecting qubits only though measurements is a defining constraint of quantum physics
- That is why in the real world you cannot discriminate these two processes!
- Do deterministic tests solve this problem?

Deterministic Tests

Non-Determinism only in the Processes

Definition (Deterministic Tests)
Let $\mathbb{T}_D \subsetneq \mathbb{T}_G$ be the set of deterministic tests, i.e. those that do not contain occurrences of the non-deterministic sum.

- They solve the counterexample presented before

$$C_{01} \not\sim_{\mathbb{T}_G} C_{\pm} \qquad\qquad C_{01} \sim_{\mathbb{T}_D} C_{\pm}$$

- The result can be generalized: deterministic tests do not distinguish distributions of states that behave the same according to quantum theory!

Lifting Indistinguishablity from Quantum Physics to lqCCS

Fact

Two distributions of quantum states $\Delta = \sum_i p_i \bullet |\psi_i\rangle$ *and* $\Theta = \sum_j q_j \bullet |\phi_j\rangle$ *are indistinguishable, written* $\Delta \cong \Theta$*, if*

$$\sum_{i \in I} p_i \cdot |\psi_i\rangle\langle\psi_i| = \sum_{j \in J} q_j \cdot |\phi_j\rangle\langle\phi_j|$$

Theorem

Given two distributions of quantum states $\Delta = \sum_i p_i \bullet |\psi_i\rangle$ *and* $\Theta = \sum_j q_j \bullet |\phi_j\rangle$ *such that* $\Delta \cong \Theta$*, it holds that for any deterministic test* $T \in \mathbb{T}_D$ *and resolution* R

$$\sum_{i \in I} p_i \cdot sp_R(\langle\!\langle |\psi_i\rangle, \mathbf{0}, T \rangle\!\rangle) = \sum_{j \in J} q_j \cdot sp_R(\langle\!\langle |\phi_j\rangle, \mathbf{0}, T \rangle\!\rangle)$$

Conclusions

Recap

- Process algebras can model concurrent quantum systems as NPLTSs where probabilities depend on a quantum state
- The standard approach of testing equivalence exceeds the observational limitations prescribed by quantum theory
- Non-determinism is the cause for this mismatch with the expected equivalences
- In a nutshell, it allows you to inspect the state of a qubit without performing a measurement, hence without altering it
- Forbidding non-deterministic tests suffices for recovering the expected indistinguishability

Some Pointers

With Fabio Gadducci, Giuseppe Lomurno @UniPi, Gabriele Tedeschi @IRIF:

POPL2024 "Quantum Bisimilarity via Barbs and Contexts: Curbing the Power of Non-deterministic Observers"
Saturated bisimilarity: we need to constrain non-determinism

APLAS2024 "Quantum Bisimilarity Is a Congruence Under Physically Admissible Schedulers"
Scheduled bisimilarity: non-determinism constrained in general

WADT2024 "Reconciling Quantum Theory and Process Equivalence via Physically Admissible Schedulers"
Trace equivalence for quantum processes

Some Pointers

ISOLA2024 "Testing Quantum Processes"

Testing equivalence for quantum processes (this lesson)

CONCUR2024 "Effect Semantics for Quantum Process Calculi"

Alternative, purely quantum model (pLTS $\rightarrow$ qLTS)

ACT2024 "A Coalgebraic Model of Quantum Bisimulation"

Same as before but using Coalgebras (with Elena Di Lavore @Tallinn)

Most recent work on Arxiv:

"Verification of Quantum Protocols Adopting Physically Admissible Schedulers"